

Task 1 Report — ML Classification Project

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1. Problem Statement

The goal of this task was to build and evaluate a supervised binary classification model to predict whether a Titanic passenger survived or not, based on features such as age, gender, ticket class, and fare.

2. Dataset & Preprocessing

Dataset: Titanic (loaded via Seaborn — 891 rows, 8 features)

Preprocessing steps performed:

- Filled missing age values with the median age (28.0)
 - Filled missing embarked values with the most frequent port ('S')
 - Encoded categorical columns (sex, embarked) using Label Encoding
 - Applied StandardScaler to normalize features for Logistic Regression
 - Applied Stratified Train/Test split (80% train, 20% test) to preserve class balance
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3. Models Used

Logistic Regression — a linear model that estimates the probability of survival using a sigmoid function. Simple, interpretable, and works well when the decision boundary is roughly linear.

Random Forest — an ensemble of 100 decision trees. Each tree votes on the outcome and the majority wins. More powerful than a single tree as it reduces overfitting.

4. Results

<u>Metric</u>	<u>Logistic Regression</u>	<u>Random Forest</u>
Accuracy	79.89%	82.68%
Precision	77.97%	80.65%
Recall	66.67%	72.46%
F1 Score	71.88%	76.34%
ROC-AUC (Test)	0.8519	0.8277
ROC-AUC (5-Fold CV)	0.853 ± 0.022	0.869 ± 0.025

5. Key Findings

- **Random Forest outperforms** Logistic Regression on accuracy, precision, recall and F1 score
 - The 5-fold cross-validation ROC-AUC of Random Forest (0.869) is higher and confirms it generalises better to unseen data
 - **Top 3 most important features** identified by Random Forest: fare, sex, and age — aligning with the historical fact that women and higher-class passengers had higher survival priority
 - Logistic Regression achieved a slightly higher single-split ROC-AUC (0.852) but cross-validation confirms Random Forest is the more reliable model
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6. Conclusion

Random Forest is selected as the final model with **82.68% accuracy** and **0.869 cross-validated ROC-AUC**. The model successfully learns survival patterns from the Titanic dataset. Future improvements could include hyperparameter tuning (GridSearchCV), handling class imbalance (SMOTE), and testing additional algorithms such as XGBoost.

GitHub Repository: : <https://github.com/sarthakcoursera/InternSpark-Internship>
